

Introduction to Differential Equations (Chapter 1)

Differential Equation: An equation containing the derivatives of one or more unknown functions (or dependent variables), with respect to one or more independent variables, is said to be a differential equation (DE).

The n th-order differential equation is: $a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = g(x)$

Classification by Type: If a differential equation contains only ordinary derivatives of one or more unknown functions with respect to a single independent variable, it is said to be an **ordinary differential equation (ODE)**. An equation involving partial derivatives of one or more unknown functions of two or more independent variables is called a **partial differential equation (PDE)**.

Examples: ODE $\Rightarrow \frac{dy}{dx} + 5y = e^x$ PDE $\Rightarrow \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial x^2} = 0$

Note: Ordinary derivatives can be written using Leibniz notation $\left(\frac{dy}{dx}\right)$ or in Prime notation (y', y'') .

Classification by Order: The order of a differential equation (either ODE or PDE) is the order of the highest derivative in the equation. For example, the previous two equations are of 1st order and 2nd order respectively.

Classification by Linearity: An n th-order ordinary differential equation is said to be linear if the following conditions are satisfied:

1. The dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ are of the first degree, that is, the power of each term involving y is 1.
2. The coefficients $a_0, a_1, \dots, a_n$ of $y, y', y'', \dots, y^{(n)}$ depend at most on the independent variable x .
3. No transcendental (Trigonometric, Logarithmic, Exponential) function of dependent variable is allowed.

A nonlinear ordinary differential equation is simply one that is not linear.

- ❖ An ordinary differential equation in which the independent variable does not appear explicitly is said to be **autonomous**.
- ❖ An n th order ordinary differential equation is said to be **homogeneous**, whereas an equation with $g(x)$ identically zero.

Solution of an ODE: Any function φ , defined on an interval I and possessing at least n derivatives that are continuous on I , which when substituted into an n th-order ordinary differential equation reduces the equation to an identity, is said to be a solution of the equation on the interval.

Interval of Definition: We cannot think of a solution to an ordinary differential equation without simultaneously thinking interval. The interval I is variously called the **interval of definition** the **interval of existence**, the **interval of validity**, or the **domain of the solution** and can be an open interval (a, b) , a closed interval $[a, b]$, an infinite interval (a, ∞) , and so on.

Trivial Solution: Any solution of a DE that is identically zero on an interval I , is said to be a trivial solution.

Explicit and Implicit Solutions: A solution in which the dependent variable is expressed solely in terms of the independent variable and constants is said to be an explicit solution. The form of an explicit solution is

$$y = \varphi(x)$$

A relation $G(x, y) = 0$ is said to be an implicit solution of an ordinary differential equation on an interval I , if there exists at least one function that satisfies the relation as well as the differential equation on I .

Families of Solutions: When solving a first-order differential equation $F(x, y, y') = 0$, we usually obtain a solution containing a single arbitrary constant or parameter c . A solution containing an arbitrary constant represents a set $G(x, y, c) = 0$ of solutions called a **one-parameter family of solutions**. When solving an n th-order differential equation, $F(x, y, y', y'', \dots, y^{(n)}) = 0$, we seek an **n -parameter family of solutions** $G(x, y, c_1, c_2, \dots, c_n) = 0$. This means that a single differential equation can possess an infinite number of solutions corresponding to the unlimited number of choices for the parameter(s). A solution of a differential equation that is free of arbitrary parameters is called a **particular solution**.

- ❖ Sometimes a differential equation possesses a solution that is not a member of a family of solutions of the equation- that is, a solution that cannot be obtained by specializing any of the parameters in the family of solutions. Such an extra solution is called a **singular solution**.

Example: Find order, degree and linearity of the following differential equations.

1. $\frac{d^3y}{dx^3} - 2e^x \frac{dy}{dx} + 7y = 0$

6. $(x+2) \frac{d^6y}{dx^6} - x^3y \frac{dy}{dx} + y = 0$

2. $-7 \frac{d^6y}{dx^6} + 2x^2 \frac{d^3y}{dx^3} + y = e^x$

7. $\frac{1}{\sqrt{\frac{dy}{dx} + y}} = x$

3. $\sqrt[3]{\frac{dy}{dx} - 1} = \sqrt{\frac{d^3y}{dx^3} + y}$

8. $-x^3 \frac{d^4y}{dx^4} + 2 \left(\frac{d^2y}{dx^2} \right)^3 + y = e^{2x} + 1$

4. $y - 3(\sin y) \frac{dy}{dx} + \frac{d^3y}{dx^3} = 0$

9. $\frac{d^3y}{dx^3} + 2 \frac{dy}{dx} = \sqrt{1 - \frac{d^3y}{dx^3}}$

5. $(\cos x) \frac{d^7y}{dx^7} + 2y^3 = \sin x$

10. $-\left(\frac{dy}{dx}\right) \frac{d^3y}{dx^3} + 2y = 0$

Exercise 1.1: 1-14, 21, 23, 24, 37, 38